Section III : Pseudo Codes Based on PYQ — Solutions & Explanations

Infosys Systems Engineer (SE) | 5-6 Nov 2026 Drive | 100% Pure SE Verified

Total Questions: 7 | **Total Marks:** 7

Section: Technical Pseudo Codes -01

Problem 1 • What is the output of the following C code? #include **int main() { int num = 987; int rem; while(num != 0) { rem = num % 4; num = num / 10; } printf("%d", rem); }**

PRACTICE VARIANT

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

What is the output of the following C code? #include int main() { int num = 987; int rem; while(num != 0) { rem = num % 4; num = num / 10; } printf("%d", rem); }

A 0

C 10

D 5

Verified Answer: Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 2 • What is the output of the following C code? #include **int main() { float x = 0.0; long int y = 10; printf("%d", sizeof(y) == sizeof(x + y)); return 0; }**

PRACTICE VARIANT

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

What is the output of the following C code? #include int main() { float x = 0.0; long int y = 10; printf("%d", sizeof(y) == sizeof(x + y)); return 0; }

A 3

C 5

D 2

Verified Answer: Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 3 • Predict the output of the given pseudocode if the value of the number is 6: Read number k = 2 i = 2 while i <= number: k = k * i i = i + 1 end while write k

◯ PRACTICE VARIANT

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

Predict the output of the given pseudocode if the value of the number is 6: Read number k = 2 i = 2 while i <= number: k = k * i i = i + 1 end while write k

B 1340.0

C 1700.0

D 1560.0

✓ **Verified Answer:** Option (A) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (A)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 4 • What is the output of the following C code? int f(int x, int *py, int **ppz) { int y, z; **ppz += 1; z = **ppz; *py += 2; y = *py; x += 3; return x + y + z; } void main() { int c, *b, **a; c = 4; b = &c; a = &b; printf("%d ", f(c, b, a)); return 0; }

◯ PRACTICE VARIANT

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

What is the output of the following C code? int f(int x, int *py, int **ppz) { int y, z; **ppz += 1; z = **ppz; *py += 2; y = *py; x += 3; return x + y + z; } void main() { int c, *b, **a; c = 4; b = &c; a = &b; printf("%d ", f(c, b, a)); return 0; }

A 18

C 21

D 22

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 5 • What is the output of the following C code? #include int main() { int a = 12; void *ptr = (int *)&a; printf("%d", *ptr); getchar(); return 0; }

◯ PRACTICE VARIANT

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

What is the output of the following C code? #include int main() { int a = 12; void *ptr = (int *)&a; printf("%d", *ptr); getchar(); return 0; }

A 12

C Run Time Error

D 0

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 6 • Predict the output of the following pseudocode. Integer x, y, z Set x = -2, y = 3, z = 1 if (x + (2 & 2) && y + (3 & 3) && z + (2 ^ 2)) x = x - 2 y = x Else x = z y = y ^ 2 End if Print x + y + z

○ PRACTICE VARIANT

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

Predict the output of the following pseudocode. Integer x, y, z Set x = -2, y = 3, z = 1 if (x + (2 & 2) && y + (3 & 3) && z + (2 ^ 2)) x = x - 2 y = x Else x = z y = y ^ 2 End if Print x + y + z

A 2

B 11

D 0

✓ **Verified Answer:** Option (C) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (C)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 7 • int main() { int x, y = 5, z = 5; x = y == z; Print(x) return 0; }

○ PRACTICE VARIANT

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

int main() { int x, y = 5, z = 5; x = y == z; Print(x) return 0; }

A 0

C 5

D Compilation Error

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.